

BUSINFO 702 | Q3 2026 | Project Report

# **Tourism Resilience Across New Zealand**

Does domestic demand stabilize off-season tourism spending?

## **Group 20**

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## 1. Executive Summary

Tourism expenditure in New Zealand reached NZ\$46.6 billion in the year ended March 2025, the highest level on record (Stats NZ, 2026). The sector also contributed 7.7% of GDP directly and indirectly (Tourism New Zealand, 2026). However, these national figures mask substantial regional differences. Tourism demand is highly seasonal, and the strength of that seasonality varies considerably across regions (Ministry of Business, Innovation and Employment [MBIE], n.d.-a).

This project examines whether domestic tourism helps offset periods of weaker international demand and identifies which Regional Tourism Organisation (RTO) areas appear most exposed to tourism shocks.

To address these questions, a SQLite data warehouse was developed by integrating three MBIE tourism datasets: Monthly Regional Tourism Estimates (MRTE) for spending, the Accommodation Data Programme (ADP) for guest nights, and Tourism Volumes and Flows (TV&F) for visitor counts. These datasets were standardised using the Stats NZ Regional Tourism Organisation Areas 2025 classification (Stats NZ, 2025). The final warehouse contains 2,259 monthly fact records across 38 RTO areas.

### Three main findings emerged:

- **Seasonality varies substantially across regions.** South Island tourism spending is more seasonal overall, ranging from about 26% above its average month in December to 27% below in June, a 53-percentage-point spread. The North Island shows a smaller spread of about 34 percentage points.
- **Domestic tourism provides only a limited off-season buffer.** Domestic visitor numbers decline in every RTO during periods of weaker international demand. However, domestic spending is more resilient and increases in 11 of the 31 RTOs with spending data, most notably in Ruapehu, where it more than doubles.
- **Potential shock exposure is concentrated in the South Island.** Eight of the ten highest-ranked RTOs on the project's vulnerability measure are located in the South Island, reflecting stronger dependence on international spending, peak-season demand, or both.

Overall, the results suggest that resilience depends less on domestic tourism in general and more on whether a destination has demand drivers that operate in different seasons. Regions with both strong international dependence and weak domestic offset appear especially exposed, meaning these risks may reinforce rather than counterbalance each other.

## 2. Research Questions

A national spending figure could tell us little about regional risk. Two destinations can report the same annual total while facing very different patterns. For example, one might earn two-thirds of its income from overseas visitors over three summer months, while the other might earn a steady domestic income year-round. The first is fragile, and the second is not, yet the annual totals look identical.

This motivates the project's overarching question: which regions sustain tourism income year-round, and does domestic demand serve as an off-season buffer that stabilizes them? Three sub-questions break this down.

### ***RQ1 - How does tourism seasonality vary across RTOs, and between the North and South Islands?***

Tourism demand in New Zealand is highly seasonal, but the strength of seasonality varies across destinations. Raw spending levels alone cannot show whether a region is more seasonal or simply has a larger tourism market. This research question therefore compares each RTO's monthly spending with its own average and applies the same approach to the North and South Islands, allowing fair comparison across areas of different sizes. The aim is to identify where seasonal fluctuations are strongest and whether broader island-level patterns exist. Highly seasonal destinations may face greater pressure during peak periods and weaker demand during quieter months.

### ***RQ2 - During months of lower international visitor demand, how do domestic visitor numbers, spending, and guest nights change across RTOs?***

Once we know which regions are unstable, the next question is whether domestic demand fills the gap. Most off-season tourism strategies assume it does, yet that assumption is rarely tested with regional data. Weak international months fall at different times in different places, so the test must run against each RTO's own demand cycle. Measuring visitors, spending, and guest nights together also separates two very different outcomes: more people arriving, and more money being spent.

### ***RQ3 - Which RTOs are most dependent on peak-season or international demand, and therefore more vulnerable to shocks?***

If domestic demand does not reliably fill the off-season gap, the final question is who carries the most risk. Vulnerability has two separate sources: reliance on overseas markets and concentration of income into a few months. A region can be exposed through either one. Identifying which regions carry both connects the analysis to the theme of a sustainable and equitable future, because shocks fall hardest on small communities with the fewest employment alternatives.

### 3. Star Schema Design

The star schema integrates the three tourism datasets into a consistent model for monthly tourism analysis by region and visitor type. The central fact table, FactTourismMonthly, has a grain of one record per Month  $\times$  RTO  $\times$  Visitor Type and stores SpendMillion, GuestNights, and VisitorCount. Descriptive attributes are separated into DimDate, DimLocation, and DimVisitorType, enabling consistent analysis across time, geography, and visitor type.

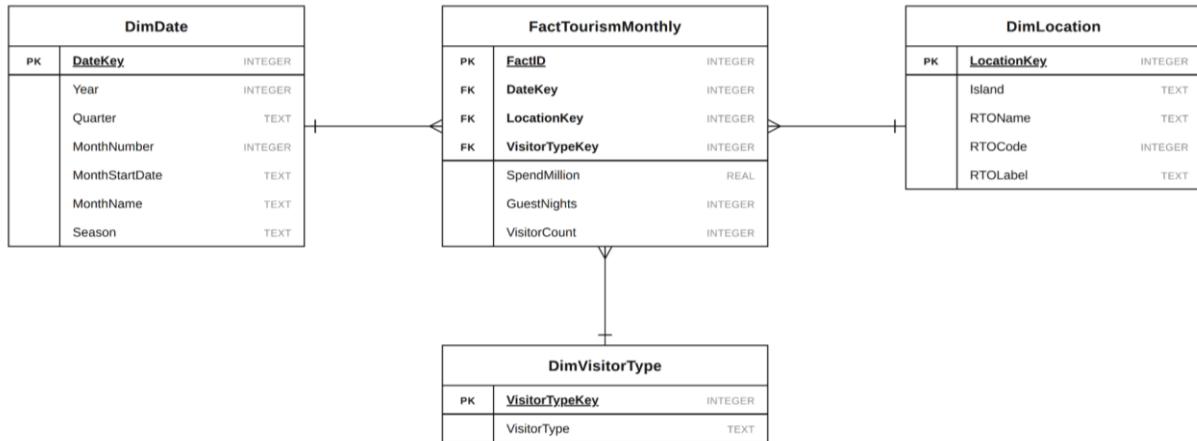


Figure 1. Star schema, drawn in draw.io.

*FactTourismMonthly sits at Month  $\times$  RTO  $\times$  Visitor Type and joins to three dimensions on integer primary keys.*

#### 3.1 How the datasets relate to the schema

The source datasets contribute complementary measures to the fact table. Each tourism dataset is standardised and aggregated to the shared fact grain before loading, while the Stats NZ RTO reference dataset supplies DimLocation attributes. Table 1 summarises these contributions.

| Source dataset                                            | Contributes            | Aggregated across                     |
|-----------------------------------------------------------|------------------------|---------------------------------------|
| <a href="#">MRTE — Monthly Regional Tourism Estimates</a> | SpendMillion           | 7 products $\times$ 29 origin markets |
| <a href="#">ADP — Accommodation Data Programme</a>        | GuestNights            | 6 accommodation property types        |
| <a href="#">TV&amp;F — Tourism Volumes and Flows</a>      | VisitorCount           | already at the fact grain             |
| <a href="#">Stats NZ RTO Areas 2025</a>                   | DimLocation attributes | reference data, 41 areas              |

Table 1. Contribution of each source dataset to the fact table.

The three tourism datasets share Month, RTO, and a visitor-type concept, but use different labels. MRTE uses 'New Zealand' and 'International'; ADP uses 'Domestic guest nights' and 'International guest nights'; and TV&F uses 'Domestic visitor' and 'Total international visitor'. These labels are standardised during cleaning, so the

sources align to the same Month  $\times$  RTO  $\times$  Visitor Type grain. MRTE product and origin are summed into SpendMillion, while ADP property type is summed into GuestNights; TV&F is already at the fact grain. The Stats NZ RTO Areas 2025 dataset provides reference attributes for DimLocation rather than a fact measure.

### 3.2 Dimensions and attribute hierarchies

The three dimensions provide descriptive context for analysing the fact measures by time, geography, and visitor type. **DimDate** supports calendar and seasonal analysis, **DimLocation** supports geographic roll-up from island to RTO, and **DimVisitorType** standardises the two visitor categories. Table 2 summarises the principal hierarchies.

| Dimension      | Hierarchy                                      | Levels                                                    |
|----------------|------------------------------------------------|-----------------------------------------------------------|
| DimDate        | Year $\rightarrow$ Quarter $\rightarrow$ Month | 3 years $\rightarrow$ 10 quarters $\rightarrow$ 30 months |
| DimLocation    | Island $\rightarrow$ RTO                       | 2 islands $\rightarrow$ 40 RTO areas                      |
| DimVisitorType | Domestic/International                         |                                                           |

Table 2. Attribute hierarchies in the dimensional model.

DimDate supports two-time roll-ups: Year  $\rightarrow$  Quarter  $\rightarrow$  Month for monthly and quarterly analysis, and Year  $\rightarrow$  Season  $\rightarrow$  Month for the southern-hemisphere seasonality analysis in Section 5. DimLocation contains the Island  $\rightarrow$  RTO hierarchy, while DimVisitorType is a two-row dimension containing Domestic and International. Storing visitor type as a dimension provides a controlled value for joins and filtering.

The Island attribute in DimLocation was assigned from Stats NZ RTO code ranges: 1–23 for North Island, 24–40 for South Island, and 99 for Area Outside RTO. The classification was checked against an independent MBIE source, which uses the same North Island and South Island grouping (MBIE, n.d.). The source reference contains 41 areas overall; Table 2 represents the 40 RTO areas, with Area Outside RTO retained as the additional reference category.

### 3.3 Naming, keys, and data types

The model follows standard dimensional-modelling conventions so that the schema is readable and enforceable without relying on external documentation.

- Tables use the Dim and Fact prefixes. Column names are singular, unabbreviated, and consistent across tables, such as DateKey, LocationKey, and VisitorTypeKey.
- Every dimension has an INTEGER primary key. The fact table stores only these keys and their measures, so no descriptive text repeats across 2,259 rows, and an RTO can be renamed without touching the fact table.

- All three foreign keys are declared with REFERENCES constraints, and PRAGMA foreign\_keys is set to ON, so an orphan fact row cannot be inserted.
- Data types follow SQLite affinities. INTEGER is used for keys, counts, and guest nights, REAL for SpendMillion, and TEXT for names, ISO-format dates, and coded attributes.
- A UNIQUE constraint on DateKey, LocationKey, and VisitorTypeKey enforces the declared grain in the database itself rather than by convention.

These conventions are visible in the fact table definition, which is reproduced with the rest of the build in Section 4.6:

```
CREATE TABLE FactTourismMonthly (  
  FactID          INTEGER PRIMARY KEY AUTOINCREMENT,  
  DateKey         INTEGER NOT NULL,  
  LocationKey     INTEGER NOT NULL,  
  VisitorTypeKey  INTEGER NOT NULL,  
  SpendMillion    REAL,  
  GuestNights     INTEGER,  
  VisitorCount    INTEGER,  
  FOREIGN KEY (DateKey)      REFERENCES DimDate(DateKey),  
  FOREIGN KEY (LocationKey)  REFERENCES DimLocation(LocationKey),  
  FOREIGN KEY (VisitorTypeKey) REFERENCES DimVisitorType(VisitorTypeKey),  
  UNIQUE (DateKey, LocationKey, VisitorTypeKey) );
```

## 4. ELT Implementation in SQLite

The project follows an ELT sequence. The raw files are loaded into SQLite, and every cleaning, conformance, and modelling step is then carried out in SQL inside the database. This section reproduces all SQLite commands and SQL statements used to build the warehouse. The three analytical queries that read the finished warehouse are reproduced in full below.

### 4.1 Extract

Four public datasets were downloaded from MBIE's Tourism Evidence and Insights Centre (MBIE, 2026a, 2026b, 2026c) and the Stats NZ Geographic Data Service (Stats NZ, 2025). The two XLSX workbooks were exported to CSV so that all four sources shared a single import format.

| Dataset                | Source                   | Raw format | Rows extracted |
|------------------------|--------------------------|------------|----------------|
| MRTE — spending by RTO | teic.mbie.govt.nz        | XLSX → CSV | 240,180        |
| ADP — accommodation    | teic.mbie.govt.nz        | CSV        | 420,736        |
| TV&F — visitor volumes | teic.mbie.govt.nz        | XLSX → CSV | 5,499          |
| RTO Areas 2025         | datafinder.stats.govt.nz | XLSX → CSV | 41             |

Table 3. Source datasets and volumes at extraction.

Extraction was limited to the common analytical period of January 2024 to June 2026, and to RTO-level records.

### 4.2 Load

Each CSV was imported as a raw table using DB Browser for SQLite. The first row supplied the column names, and every field was loaded as TEXT so that no value could be silently coerced or truncated during import. The cleaning script then clears any earlier versions of the four clean tables so it can be re-run from scratch. Foreign key enforcement is switched off for the drops, because RTO\_Areas\_Mapping\_Clean is referenced by the other three, and switched back on immediately afterward.

```
PRAGMA foreign_keys = OFF;

DROP TABLE IF EXISTS Visitor_Count_Clean;
DROP TABLE IF EXISTS ADP_Nights_Clean;
DROP TABLE IF EXISTS Spend_Clean;
DROP TABLE IF EXISTS RTO_Areas_Mapping_Clean;

PRAGMA foreign_keys = ON;
```

### 4.3 Transform: cleaning the raw tables

The four raw tables were then converted into typed, constrained, clean tables. Each clean table is created with its constraints first, then populated by a single INSERT ... SELECT that reads from the raw table. The RTO mapping was built first, because the three fact sources are all foreign-keyed to it.

#### 4.3.1 The RTO mapping

RTO\_Areas\_Mapping\_Clean contains the official Stats NZ code and the four name variants used across the three sources. TRIM removes stray whitespace, and NULLIF converts the literal string 'null' produced by the workbook export into a real NULL. The UNIQUE constraints on the two-name columns guarantee that the joins in the three statements that follow cannot match more than one row.

```
CREATE TABLE RTO_Areas_Mapping_Clean (
  RTO_Code          INTEGER PRIMARY KEY,
  RTO2025_V1_00_NAME TEXT NOT NULL UNIQUE,
  RTO2025_V1_00_NAME_ASCII TEXT,
  RTO_Label_2025    TEXT,
  RTO_Label_2025_ASCII TEXT UNIQUE
);

/* Insert cleaned RTO mapping data */

INSERT INTO RTO_Areas_Mapping_Clean (
  RTO_Code,
  RTO2025_V1_00_NAME,
  RTO2025_V1_00_NAME_ASCII,
  RTO_Label_2025,
  RTO_Label_2025_ASCII
)
SELECT
  CAST(RTO2025_V1_00 AS INTEGER),
  TRIM(RTO2025_V1_00_NAME),
  NULLIF(
    TRIM(RTO2025_V1_00_NAME_ASCII),
    'null'
  ),
  NULLIF(
    TRIM(RTO_Label_2025),
    'null'
  ),
  NULLIF(
    TRIM(RTO_Label_2025_ASCII),
    'null'
  )
FROM "Raw-RTO Areas Mapping";
```

#### 4.3.2 Visitor counts (TV&F)

Dates arrived in two different formats across the sources, so all were normalized to the first day of the month. This gives every source a single TEXT join key. The CASE expression is also applied identically in all cleaning statements. The RTO code is derived from the last two characters of the destination code, and the two population segments are renamed to the standard Domestic and International labels.

```
CREATE TABLE Visitor_Count_Clean (
  Visitor_Count_ID INTEGER PRIMARY KEY,
  RTO_Code INTEGER NOT NULL,
  Destination TEXT NOT NULL,
  Visitor_type TEXT NOT NULL
  CHECK (
    Visitor_type IN (
      'Domestic',
      'International'
    )
  ),
  Month TEXT NOT NULL,
  Visitor_count INTEGER NOT NULL
  CHECK (Visitor_count >= 0),
  FOREIGN KEY (RTO_Code)
  REFERENCES RTO_Areas_Mapping_Clean(RTO_Code),
  UNIQUE (
    RTO_Code,
    Visitor_type,
    Month
  )
);

/* Insert cleaned visitor-count data */

INSERT INTO Visitor_Count_Clean (
  RTO_Code,
  Destination,
  Visitor_type,
  Month,
  Visitor_count
)
SELECT
  CAST(
    SUBSTR(
      TRIM(Destination_code),
      -2
    ) AS INTEGER
  ) AS RTO_Code,
  TRIM(
    REPLACE(
      Destination,
      ' RTO',
      ''
    )
  ) AS Destination,
  CASE
```

```

        WHEN TRIM(Population_segment) =
            'Domestic visitor'
        THEN 'Domestic'

        WHEN TRIM(Population_segment) =
            'Total international visitor'
        THEN 'International'
    END AS Visitor_type,

    /* Supports DD/MM/YYYY and YYYY-MM-DD */
    CASE
        WHEN INSTR(TRIM(Date), '/') > 0
        THEN DATE(
            SUBSTR(TRIM(Date), -4)
            || '-'
            || PRINTF(
                '%02d',
                CAST(
                    SUBSTR(
                        TRIM(Date),
                        INSTR(TRIM(Date), '/') + 1,
                        INSTR(
                            SUBSTR(
                                TRIM(Date),
                                INSTR(TRIM(Date), '/') + 1
                            ),
                            '/'
                        ) - 1
                    ) AS INTEGER
                )
            || '-01'
        )

        ELSE DATE(Date, 'start of month')
    END AS Month,

    CAST(
        Monthly_unique_counts
        AS INTEGER
    ) AS Visitor_count

FROM Visitor_Count_Raw

WHERE TRIM(Geographic_level_destination) = 'RTO'

AND TRIM(Population_segment) IN (
    'Domestic visitor',
    'Total international visitor'
);

```

### 4.3.3 Accommodation nights (ADP)

ADP values suppressed for confidentiality, flagged 'c' by MBIE, were excluded rather than read as zero, because a suppressed value means unknown rather than none. The same WHERE clause also removes the national and all-property totals that ADP publishes alongside the detail rows. ADP records the area name with an ' RTO' suffix, which the ADP\_Standardised CTE strips before joining on the official Stats NZ name.

```
CREATE TABLE ADP_Nights_Clean (
  ADP_Nights_ID INTEGER PRIMARY KEY,

  RTO_Code INTEGER NOT NULL,

  Month TEXT NOT NULL,

  Property TEXT NOT NULL,

  Visitor_type TEXT NOT NULL
  CHECK (
    Visitor_type IN (
      'Domestic',
      'International'
    )
  ),

  Total_stay_nights INTEGER NOT NULL
  CHECK (Total_stay_nights >= 0),

  FOREIGN KEY (RTO_Code)
    REFERENCES RTO_Areas_Mapping_Clean(RTO_Code),

  UNIQUE (
    RTO_Code,
    Month,
    Property,
    Visitor_type
  )
);

/* Insert cleaned accommodation data */

INSERT INTO ADP_Nights_Clean (
  RTO_Code,
  Month,
  Property,
  Visitor_type,
  Total_stay_nights
)

WITH ADP_Standardised AS (
  SELECT
    /* Convert DD/MM/YYYY into YYYY-MM-01 */
    CASE
      WHEN INSTR(TRIM(Month), '/') > 0
      THEN DATE(
        SUBSTR(TRIM(Month), -4)
        || '-'
        || PRINTF(
```

```

        '%02d',
        CAST(
            SUBSTR(
                TRIM(Month),
                INSTR(TRIM(Month), '/') + 1,
                INSTR(
                    SUBSTR(
                        TRIM(Month),
                        INSTR(TRIM(Month), '/') + 1
                    ),
                    '/',
                ) - 1
            ) AS INTEGER
        )
    ) || '-01'
)

ELSE DATE(Month, 'start of month')
END AS Clean_Month,

TRIM(
CASE
    WHEN TRIM(Area) LIKE '% RTO'
    THEN SUBSTR(
        TRIM(Area),
        1,
        LENGTH(TRIM(Area)) - 4
    )
    ELSE TRIM(Area)
END
) AS Clean_Area,

TRIM(Property) AS Property,

CASE
    WHEN TRIM(Measure) =
        'Domestic guest nights'
    THEN 'Domestic'

    WHEN TRIM(Measure) =
        'International guest nights'
    THEN 'International'
END AS Visitor_type,

CAST(Value AS INTEGER)
AS Total_stay_nights

FROM "Raw-ADP(nights)"

WHERE TRIM("Area type") = 'RTO'

AND TRIM(Area) <> 'Total New Zealand'

AND TRIM(Property) <> 'Total'

AND TRIM(Measure) IN (
    'Domestic guest nights',
    'International guest nights'
)

```

```

    )
    AND COALESCE(
        TRIM(Flag),
        ''
    ) <> 'c'
)

```

#### 4.3.4 Tourism spending (MRTE)

Non-RTO spending areas were removed from MRTE to prevent regional totals from being inflated by activity that belongs to no RTO. The second condition in the same clause is where MRTE's 'New Zealand' label is restricted before being renamed 'Domestic'. MRTE records the organization's trading name rather than the official area name, so it joins on the ASCII label instead.

```

CREATE TABLE Spend_Clean (
    Spend_ID INTEGER PRIMARY KEY,
    RTO_Code INTEGER NOT NULL,
    Month TEXT NOT NULL,
    Product TEXT NOT NULL,
    Visitor_type TEXT NOT NULL
    CHECK (
        Visitor_type IN (
            'Domestic',
            'International'
        )
    ),
    Origin TEXT NOT NULL,
    Annual_Spend_Million REAL,
    Monthly_Spend_Million REAL,

    FOREIGN KEY (RTO_Code)
        REFERENCES RTO_Areas_Mapping_Clean(RTO_Code),

    UNIQUE (
        RTO_Code,
        Month,
        Product,
        Visitor_type,
        Origin
    )
);

/* Insert cleaned tourism-spending data */

INSERT INTO Spend_Clean (
    RTO_Code,
    Month,
    Product,
    Visitor_type,
    Origin,
    Annual_Spend_Million,
    Monthly_Spend_Million
)

WITH Spend_Standardised AS (

```

```

SELECT
  /* Convert DD/MM/YYYY into YYYY-MM-01 */
  CASE
    WHEN INSTR(TRIM(Date), '/') > 0
    THEN DATE(
      SUBSTR(TRIM(Date), -4)
      || '-'
      || PRINTF(
        '%02d',
        CAST(
          SUBSTR(
            TRIM(Date),
            INSTR(TRIM(Date), '/') + 1,
            INSTR(
              SUBSTR(
                TRIM(Date),
                INSTR(TRIM(Date), '/') + 1
              ),
              '/',
            ) - 1
          ) AS INTEGER
        )
      )
      || '-01'
    )
    ELSE DATE(Date, 'start of month')
  END AS Clean_Month,

  TRIM(RTO) AS Clean_RTO,

  TRIM(Product) AS Product,

  CASE
    WHEN TRIM("Visitor Type") =
      'New Zealand'
    THEN 'Domestic'

    WHEN TRIM("Visitor Type") =
      'International'
    THEN 'International'
  END AS Visitor_type,

  TRIM(Origin) AS Origin,

  CAST(
    TRIM("Annual Spend")
    AS REAL
  ) AS Annual_Spend_Million,

  CAST(
    TRIM("Monthly Spend")
    AS REAL
  ) AS Monthly_Spend_Million

FROM Raw_Spend

WHERE TRIM(RTO) NOT IN (
  'Area Outside RTO',

```

```

        'non-RTO - Horowhenua',
        'non-RTO - Kawerau',
        'non-RTO - Otorohanga',
        'non-RTO - Rangitikei',
        'non-RTO - South Waikato',
        'non-RTO - Tararua',
        'non-RTO - Waimate',
        'non-RTO - Waitomo',
        'non-RTO - Whakatane'
    )

    AND TRIM("Visitor Type") IN (
        'New Zealand',
        'International'
    )
)

```

#### ***4.3.5 Join keys, and dropping the raw tables***

Linking the three sources to the RTO mapping; therefore, required a different key in each case, as the three statements above show:

- TV&F supplies a destination code, so the RTO code is cast from its last two characters.
- ADP joins on RTO2025\_V1\_00\_NAME, the official Stats NZ area name.
- MRTE joins on RTO\_Label\_2025\_ASCII, the organisation's trading name.

Each of the three is an inner join, so any label that fails to match is dropped rather than loaded against a null region. Once the four clean tables had been verified, the raw tables were dropped so that the delivered database remains compact.

## 4.4 Transform: Time Dimension - DimDate

The warehouse script opens by enforcing foreign keys and clearing any earlier version of the star schema, in reverse dependency order so that the fact table is removed before the dimensions it references.

```
PRAGMA foreign_keys = ON;

DROP TABLE IF EXISTS FactTourismMonthly;
DROP TABLE IF EXISTS DimVisitorType;
DROP TABLE IF EXISTS DimLocation;
DROP TABLE IF EXISTS DimDate;
```

**DimDate** is built as a continuous monthly calendar rather than from the dates that happen to appear in the data. The dimension first finds the earliest and latest month across all three cleaned sources, and a recursive CTE then generates every month between them. Populating it from the distinct observed dates instead would make a missing month invisible, because the dimension and the data would agree by construction. Generating the calendar independently is what allows the completeness check in Section 4.7 to detect a month that no source reports.

The Year, Quarter and Month hierarchy and the New Zealand season are derived from the generated month. CHECK constraints on MonthNumber, Quarter and Season make an invalid value impossible to insert.

```
CREATE TABLE DimDate (

    DateKey INTEGER PRIMARY KEY,
    MonthStartDate TEXT NOT NULL UNIQUE,
    MonthNumber INTEGER NOT NULL
        CHECK (
            MonthNumber BETWEEN 1 AND 12
        ),

    MonthName TEXT NOT NULL,
    Quarter TEXT NOT NULL
        CHECK (
            Quarter IN (
                'Q1',
                'Q2',
                'Q3',
                'Q4'
            )
        ),

    Year INTEGER NOT NULL,
    Season TEXT NOT NULL
        CHECK (
            Season IN (
                'Summer',
                'Autumn',
                'Winter',
                'Spring'
            )
        )
);
```

```

/* -----
Generate the continuous monthly series.

DateBounds finds the earliest and latest month across all
three cleaned datasets.

MonthSeries then recursively creates every month between
these two dates.
----- */

```

WITH RECURSIVE

DateBounds AS (

```

    SELECT
        MIN(Month) AS MinMonth,
        MAX(Month) AS MaxMonth

```

FROM (

```

    SELECT Month
    FROM Spend_Clean

```

UNION ALL

```

    SELECT Month
    FROM ADP_Nights_Clean

```

UNION ALL

```

    SELECT Month
    FROM Visitor_Count_Clean

```

)

),

MonthSeries(MonthStartDate) AS (

```

    /* First calendar month */
    SELECT MinMonth
    FROM DateBounds

```

UNION ALL

/\* Add one month until the maximum date is reached \*/

```

SELECT
    date(
        MonthStartDate,
        '+1 month'
    )

```

FROM

```

    MonthSeries,
    DateBounds

```

WHERE

```

    MonthStartDate < MaxMonth

```

)

```

INSERT INTO DimDate (

    DateKey,
    MonthStartDate,
    MonthNumber,
    MonthName,
    Quarter,
    Year,
    Season

)

SELECT

    /* Example:
       January 2025 -> 202501
    */
    CAST(
        strftime(
            '%Y',
            MonthStartDate
        ) AS INTEGER
    ) * 100

    +

    CAST(
        strftime(
            '%m',
            MonthStartDate
        ) AS INTEGER
    )

    AS DateKey,

    MonthStartDate,

    CAST(
        strftime(
            '%m',
            MonthStartDate
        ) AS INTEGER
    )

    AS MonthNumber,

    CASE strftime('%m', MonthStartDate)

        WHEN '01' THEN 'January'
        WHEN '02' THEN 'February'
        WHEN '03' THEN 'March'
        WHEN '04' THEN 'April'
        WHEN '05' THEN 'May'
        WHEN '06' THEN 'June'
        WHEN '07' THEN 'July'
        WHEN '08' THEN 'August'
        WHEN '09' THEN 'September'
        WHEN '10' THEN 'October'
        WHEN '11' THEN 'November'
    
```

```

        WHEN '12' THEN 'December'

END AS MonthName,

CASE

    WHEN CAST(
        strftime('%m', MonthStartDate)
        AS INTEGER
    ) BETWEEN 1 AND 3
        THEN 'Q1'

    WHEN CAST(
        strftime('%m', MonthStartDate)
        AS INTEGER
    ) BETWEEN 4 AND 6
        THEN 'Q2'

    WHEN CAST(
        strftime('%m', MonthStartDate)
        AS INTEGER
    ) BETWEEN 7 AND 9
        THEN 'Q3'

    ELSE 'Q4'

END AS Quarter,

CAST(
    strftime(
        '%Y',
        MonthStartDate
    ) AS INTEGER
)

AS Year,

/* NZ seasons */
CASE

    WHEN strftime(
        '%m',
        MonthStartDate
    ) IN (
        '12',
        '01',
        '02'
    )
        THEN 'Summer'

    WHEN strftime(
        '%m',
        MonthStartDate
    ) IN (
        '03',
        '04',
        '05'
    )
        THEN 'Autumn'

```

```
        WHEN strftime(
            '%m',
            MonthStartDate
        ) IN (
            '06',
            '07',
            '08'
        )
        THEN 'Winter'

        ELSE 'Spring'

    END AS Season

FROM MonthSeries

ORDER BY MonthStartDate;
```

The primary key is an integer in YYYYMM form, which gives the dimension a compact natural sort order. The result is 30 rows covering January 2024 to June 2026, spanning three calendar years and ten quarters.

## 4.5 Transform: The Remaining Dimensions - DimLocation & DimVisitorType

The Island-to-RTO hierarchy is derived in **DimLocation** from the Stats NZ code ranges described in Section 3.2. The permitted values are constrained, so a typing error cannot create a third island.

```
CREATE TABLE DimLocation (
  LocationKey  INTEGER PRIMARY KEY,
  RTOCode     INTEGER NOT NULL UNIQUE,
  RTOName     TEXT NOT NULL,
  RTOLabel    TEXT,
  Island      TEXT NOT NULL
    CHECK (
      Island IN (
        'North Island',
        'South Island',
        'Other'
      )
    )
);

/*
The cleaned RTO mapping table contains RTO codes 1-40,
plus 99 for Area Outside RTO.

Codes 1-23 are North Island RTO areas.
Codes 24-40 are South Island RTO areas.
99 is Area Outside RTO.
*/

INSERT INTO DimLocation (
  LocationKey,
  RTOCode,
  RTOName,
  RTOLabel,
  Island
)
SELECT
  RTO_Code AS LocationKey,
  RTO_Code AS RTOCode,
  RTO2025_V1_00_NAME AS RTOName,
  RTO_Label_2025 AS RTOLabel,

  CASE
    WHEN RTO_Code BETWEEN 1 AND 23
      THEN 'North Island'

    WHEN RTO_Code BETWEEN 24 AND 40
      THEN 'South Island'

    ELSE 'Other'
  END AS Island

FROM RTO_Areas_Mapping_Clean
ORDER BY RTO_Code;
```

This produces 41 rows: 23 North Island RTOs, 17 South Island RTOs, and one 'Other' for Area Outside RTO, which is excluded from all analysis. **DimVisitorType** is loaded as two explicit rows with a matching CHECK constraint, so each label is stored once and the fact table carries only the key.

```
CREATE TABLE DimVisitorType (  
    VisitorTypeKey  INTEGER PRIMARY KEY,  
    VisitorType     TEXT NOT NULL UNIQUE  
        CHECK (VisitorType IN ('Domestic','International'))  
);  
  
INSERT INTO DimVisitorType (  
    VisitorTypeKey,  
    VisitorType  
)  
VALUES  
    (1, 'Domestic'),  
    (2, 'International');
```

## 4.6 Transform: The Fact Table - FactTourismMonthly

The fact table stores the three foreign keys and the three measures.

```
CREATE TABLE FactTourismMonthly (  
    FactID                INTEGER PRIMARY KEY AUTOINCREMENT,  
  
    DateKey                INTEGER NOT NULL,  
  
    LocationKey            INTEGER NOT NULL,  
  
    VisitorTypeKey         INTEGER NOT NULL,  
  
    SpendMillion           REAL,  
  
    GuestNights            INTEGER,  
  
    VisitorCount           INTEGER,  
  
    FOREIGN KEY (DateKey)  
        REFERENCES DimDate(DateKey),  
  
    FOREIGN KEY (LocationKey)  
        REFERENCES DimLocation(LocationKey),  
  
    FOREIGN KEY (VisitorTypeKey)  
        REFERENCES DimVisitorType(VisitorTypeKey),  
  
    UNIQUE (  
        DateKey,  
        LocationKey,  
        VisitorTypeKey  
    )  
);
```

The fact load step brings the three sources together, and it is built in five stages (Step A to Step E) within a single statement. **Step A** forms a skeleton for every Month, RTO, and Visitor Type combination that appears in any source. **Steps B to D** aggregate each source to that grain: MRTE across product and origin, ADP across the six property types, and TV&F, which is already at the grain. **Step E** joins the skeleton to the dimensions and left joins the three aggregates.

```
/* -----  
Step A: FactKeys with  
RTO + Month + Visitor Type  
----- */  
WITH FactKeys AS (  
  
    SELECT  
        RTO_Code,  
        Month,  
        Visitor_type  
    FROM Spend_Clean  
  
    UNION  
  
    SELECT
```

```

        RTO_Code,
        Month,
        Visitor_type
    FROM ADP_Nights_Clean

    UNION

    SELECT
        RTO_Code,
        Month,
        Visitor_type
    FROM Visitor_Count_Clean
),

/* -----
Step B:
Aggregate tourism spending to the common fact grain.

Product and Origin are deliberately removed here because
the analytical fact table is at:
    RTO + Month + Visitor Type

Monthly_Spend_Million is used rather than annual spending.
----- */

SpendAgg AS (

    SELECT
        RTO_Code,
        Month,
        Visitor_type,

        SUM(Monthly_Spend_Million) AS SpendMillion

    FROM Spend_Clean

    GROUP BY
        RTO_Code,
        Month,
        Visitor_type
),

/* -----
Step C:
Aggregate guest nights across accommodation property types.
----- */

NightsAgg AS (

    SELECT
        RTO_Code,
        Month,
        Visitor_type,

        SUM(Total_stay_nights) AS GuestNights

    FROM ADP_Nights_Clean

    GROUP BY

```

```

        RTO_Code,
        Month,
        Visitor_type
    ),

/* -----
Step D:
Standardise visitor counts to the same grain.
SUM() is used defensively even though the cleaned table
already has a uniqueness rule for RTO + visitor type + month.
----- */

VisitorsAgg AS (

    SELECT
        RTO_Code,
        Month,
        Visitor_type,

        SUM(Visitor_count) AS VisitorCount

    FROM Visitor_Count_Clean

    GROUP BY
        RTO_Code,
        Month,
        Visitor_type
)

/* -----
Step E:
Load all measures into the fact table.
----- */

INSERT INTO FactTourismMonthly (
    DateKey,
    LocationKey,
    VisitorTypeKey,
    SpendMillion,
    GuestNights,
    VisitorCount
)

SELECT

    d.DateKey,
    l.LocationKey,
    v.VisitorTypeKey,
    s.SpendMillion,
    n.GuestNights,
    vc.VisitorCount

FROM FactKeys fk

JOIN DimDate d
    ON d.MonthStartDate = fk.Month

JOIN DimLocation l
    ON l.RTOCode = fk.RTO_Code

```

```

JOIN DimVisitorType v
  ON v.VisitorType = fk.Visitor_type

LEFT JOIN SpendAgg s
  ON s.RTO_Code = fk.RTO_Code
  AND s.Month = fk.Month
  AND s.Visitor_type = fk.Visitor_type

LEFT JOIN NightsAgg n
  ON n.RTO_Code = fk.RTO_Code
  AND n.Month = fk.Month
  AND n.Visitor_type = fk.Visitor_type

LEFT JOIN VisitorsAgg vc
  ON vc.RTO_Code = fk.RTO_Code
  AND vc.Month = fk.Month
  AND vc.Visitor_type = fk.Visitor_type;

```

UNION is used to build the key skeleton rather than a join, because an inner join would quietly drop a valid RTO-month simply because one of the three datasets does not report it. That would bias the analysis toward the best-covered regions. The three aggregates are then left-joined, so a measure with no matching source row is stored as NULL rather than zero.

MRTE publishes both monthly and annual spending figures, but the annual figure is a rolling 12-month total, so each month already includes the previous 11 months. Summing it would count the same dollar twelve times. Therefore, only the monthly figure is loaded.

Four indexes support the dimension joins and the per-RTO grouping used throughout Section 5.

```

CREATE INDEX idx_fact_date
ON FactTourismMonthly(DateKey);

CREATE INDEX idx_fact_location
ON FactTourismMonthly(LocationKey);

CREATE INDEX idx_fact_visitor_type
ON FactTourismMonthly(VisitorTypeKey);

CREATE INDEX idx_fact_location_date
ON FactTourismMonthly(LocationKey, DateKey);

```

## 4.7 Validation

Validation runs in three stages. Section 7 of the script inspects each dimension in turn, then a sample of joined fact rows, confirming that both hierarchies are populated correctly and that a fact row resolves to readable, descriptive attributes. Sections 8 and 9 test completeness, first at month level and then at RTO-month level. Table 4 summarizes the validation results and shows the state of the delivered warehouse.

| Check                       | Result   | Interpretation                          |
|-----------------------------|----------|-----------------------------------------|
| Fact table rows             | 2,259    | 38 RTOs across 30 months                |
| RTOs represented            | 38 of 41 | 3 areas report in no source             |
| Rows missing SpendMillion   | 399      | 7 RTOs are absent from the MRTE release |
| Rows missing GuestNights    | 759      | 13 RTOs suppressed or absent from ADP   |
| Rows missing VisitorCount   | 93       | international counts end in March 2026  |
| Duplicate rows at the grain | 0        | the UNIQUE constraint holds             |
| Orphan foreign keys         | 0        | referential integrity holds             |

*Table 4. Validation results for the delivered warehouse.*

Coverage clearly differs by source, since 31 RTOs report spending, 25 report guest nights, and 38 report visitor counts. Every analytical query, therefore, filters on the relevant measure being NOT NULL, rather than assuming complete coverage. This allows each finding in the next section to specify which RTOs it applies to.

## 5. SQL for Business Analytics

Following validation of the data warehouse, each research question is addressed using a multi-stage SQL query structured with common table expressions (CTEs). **All analyses are restricted to the complete calendar years 2024 and 2025**, ensuring that each month is represented twice and allowing consistent year-on-year averaging and seasonal comparison. In the report, only the key SQL fragments that define the analytical measures and calculations are presented. The complete queries are provided in Appendix A.

### 5.1 Query 1: Tourism spending seasonality varies across RTOs and between the North and South Islands

#### *Logic*

Tourism spending was analysed over the two complete calendar years, 2024–2025. Monthly spending was first aggregated for each RTO and then averaged across the two years. A seasonality index measure was calculated by comparing each calendar month with the destination's own average monthly spending:

```
ROUND(  
    100.0  
    * AvgMonthlySpendMillion  
    / AreaAverageMonthlySpend,  
    1  
) AS SeasonalityIndex
```

A seasonality index of **100** represents the area's average monthly tourism spending across 2024–2025. Values above 100 show how far spending is above average. For example, an index of 124 means spending is 24% higher than the average month, while values below 100 indicate spending below average. Because each area is compared with its own average, RTOs and islands with very different total spending levels can be compared on the same scale.

The monthly results were then aggregated separately for the North and South Islands:

```
SELECT  
    Year,  
    MonthNumber,  
    MonthName,  
    Island,  
    SUM(TotalSpendMillion) AS TotalSpendMillion  
FROM MonthlySpend  
GROUP BY Year, MonthNumber, MonthName, Island;
```

This allows both the timing and size of seasonal peaks and troughs to be compared between the two islands.

To complement the monthly index, the standard deviation of the 12 monthly seasonality indices was calculated. A higher standard deviation indicates that spending varies more strongly throughout the year. The key calculation was:

```
ROUND(
  SQRT(
    MeanSquaredIndex
    - MeanSeasonalityIndex * MeanSeasonalityIndex), 2) AS SeasonalityStdDev
```

This was first calculated at island level and then separately for each RTO. For the RTO ranking, only destinations containing all 12 calendar months were included:

```
WHERE NumberOfMonths = 12
ORDER BY
  SeasonalityStdDev DESC
```

This prevents destinations with incomplete monthly coverage from being unfairly compared with RTOs that have a complete annual pattern.

## Result and interpretation

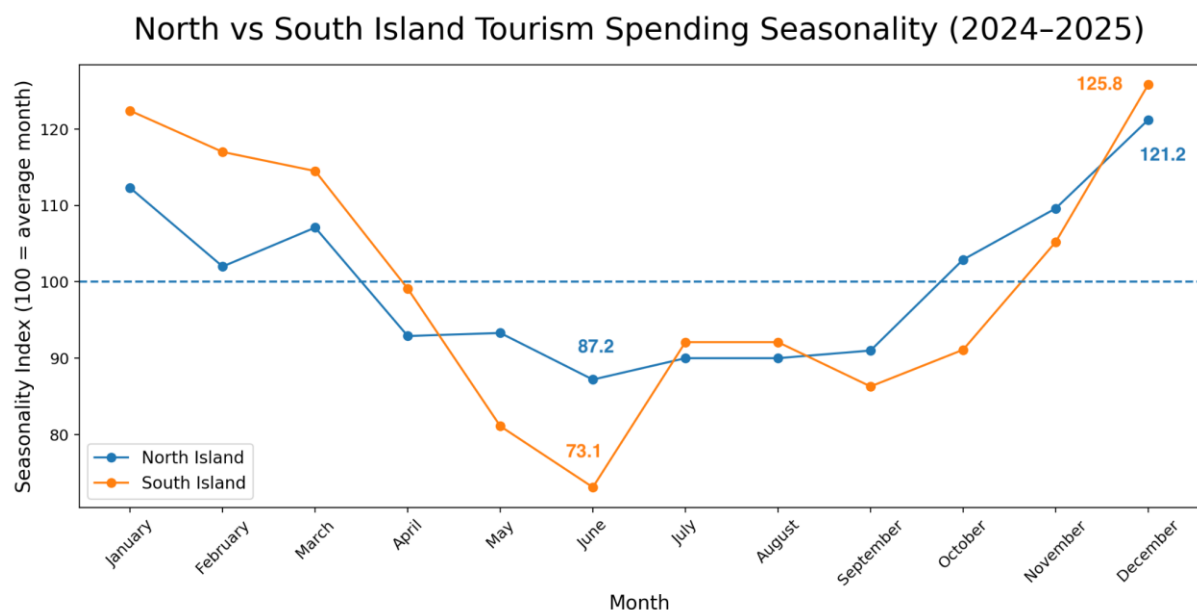
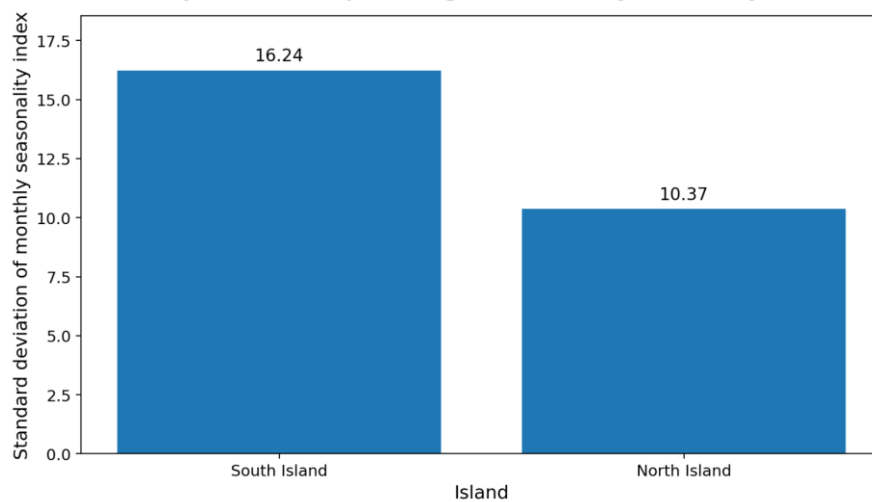


Figure 2. Seasonality index of total tourism spending by island, averaged over 2024 and 2025.

According to Figure 2, both islands show a clear summer peak and a mid-year decline, but the South Island experiences substantially stronger seasonal variation. The South Island seasonality index reaches approximately 126 in December and 122 in January, before falling to about 73 in June, representing a peak-to-trough spread of around 53 index points. The North Island varies less sharply, ranging from approximately 121 in December to 87 in June, a spread of about 34 points.

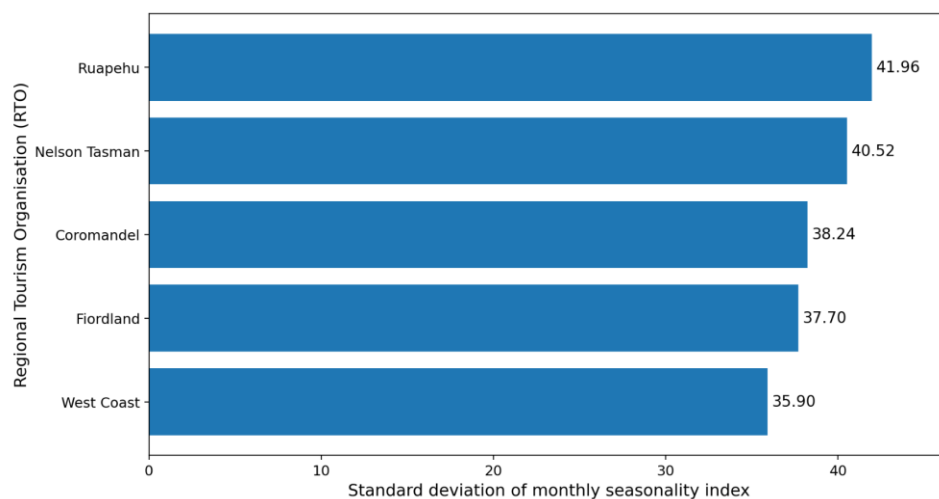
### Variation in Monthly Tourism Spending Seasonality Index by Island (2024–2025)



*Figure 3. Standard deviation of the monthly tourism spending seasonality index by island, averaged over 2024 and 2025.*

The standard deviation results confirm this difference across the full annual cycle. The South Island records a seasonality standard deviation of 16.24, compared with 10.37 for the North Island (Figure 3), showing that South Island tourism spending fluctuates more strongly around its average monthly level.

### Top 5 RTOs with the Greatest Variation in Tourism Spending Seasonality (2024–2025)



*Figure 4. Top five RTOs by standard deviation of the monthly tourism spending seasonality index, averaged over 2024 and 2025.*

Variation is even greater at the individual RTO level. As shown in Figure 4, Ruapehu records the highest standard deviation at 41.96, followed by Nelson Tasman (40.52), Coromandel (38.24) and Fiordland (37.70). These destinations therefore experience much stronger seasonal changes in tourism expenditure than the island averages. The results suggest that although the South Island is more seasonal overall, seasonal dependence is driven most strongly by the characteristics of individual destinations rather than island location alone.

**Business relevance:** The findings indicate that highly seasonal RTOs may be more dependent on a limited number of high-demand months. This can create greater exposure to tourism disruptions during peak periods and may also contribute to uneven use of tourism infrastructure and employment throughout the year. By contrast, destinations with lower seasonal variation have a more stable distribution of tourism spending and may therefore be less dependent on a narrow tourism season.

## 5.2 Query 2: Domestic response to weak international demand

As Query 1 established that seasonality varies widely between regions, Query 2 then tests whether domestic visitors fill the resulting trough.

### *Logic*

The analysis examined whether domestic tourism activity changes during months when international visitor demand is relatively weak. For each RTO, average monthly international visitor demand was first calculated:

```
SELECT
    LocationKey,
    AVG(InternationalVisitorCount) AS AvgInternationalVisitors
FROM InternationalDemand
GROUP BY LocationKey;
```

Each RTO-month was then classified as either a “Low international-demand” month or “Other months”. A month was considered low demand when the international visitor count was below that RTO's own 2024–2025 monthly average:

```
CASE
    WHEN i.InternationalVisitorCount
        < a.AvgInternationalVisitors
    THEN 'Low international demand'
    ELSE 'Other months'
END AS DemandPeriod
```

Using an RTO-specific benchmark allows destinations of different sizes to be compared fairly. For example, a low-demand month in Queenstown is determined relative to Queenstown's normal international visitor level rather than against a single national threshold.

Domestic activity in the same RTO-months was then measured using three indicators:

```
f.VisitorCount AS DomesticVisitors,
f.SpendMillion AS DomesticSpendMillion,
f.GuestNights AS DomesticGuestNights
```

The average value of each measure was calculated separately for low international-demand months and other months. Percentage differences were then calculated using the “Other months” average as the baseline as follows:

$$\text{Percentage Difference} = \frac{\text{Low-demand average} - \text{Other-month average}}{\text{Other-month average}} \times 100$$

The spending difference was calculated as:

```
ROUND(100.0 * (LowSpend - OtherSpend) / NULLIF(OtherSpend, 0), 1) AS SpendDiffPct
```

The same calculation was applied to domestic visitor numbers and guest nights. A positive percentage means the measure was higher during low international-demand months, while a negative percentage means it was lower. For example, a SpendDiffPct of **12.0** means average domestic spending is 12% higher during low international-demand months than during other months. A value of **-15.0** means it is 15% lower.

### Result and interpretation

Domestic visitor numbers fall during weak international months across all RTOs, by a median of 19.9%, and by between 12% and 46% across the full range. This result suggests New Zealanders do not arrive in greater numbers when overseas visitors leave, because both are largely summer markets and their troughs coincide.

Domestic spending behaves differently. Across the 31 RTOs with spending data the median change is only -5.6%, against -19.9% for visitor numbers, and in 11 it rises outright. Ruapehu is the clearest case, with domestic visitors down 17.6% and spending up 110.7%. Queenstown loses 23.6% of its domestic visitors but gains 10.9% in spending; Waikato, Wellington and Rotorua each gain 12% to 13%. Fewer domestic visitors arrive, and those who do often spend more.

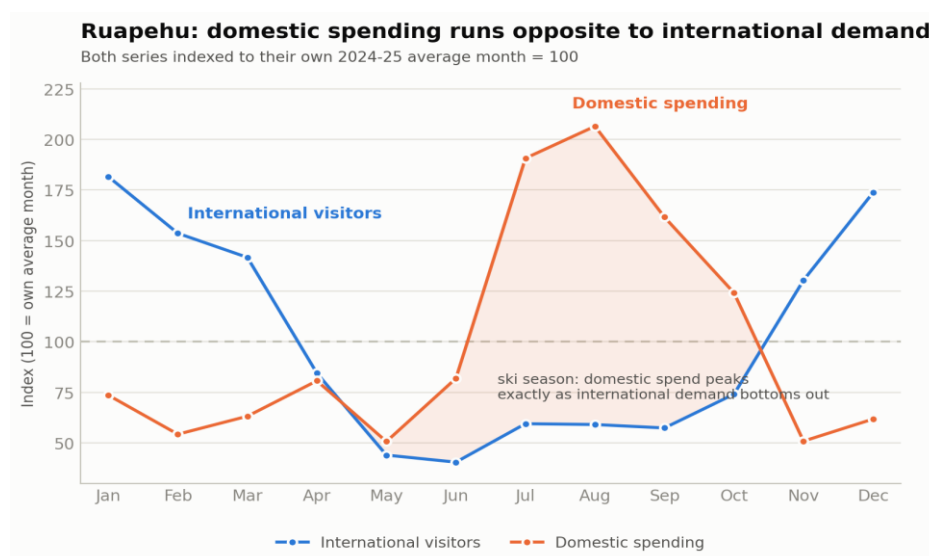


Figure 5. Ruapehu. International visitors and domestic spending, each indexed to its own 2024–25 average month.

Figure 5 shows Ruapehu's domestic spending peaks at roughly twice its average in July, August, and September, precisely when international visitors fall to around 40% of their number. This is the winter ski season, a domestic demand driver that peaks in the opposite season to international arrivals. Domestic spending averages \$11.92 million per month during Ruapehu's weak international months, compared with \$5.65 million in the others.

The wider lesson is that a domestic buffer is not automatic. It exists only where a region has a demand driver in a different season. Ruapehu has snow, while Wellington and Rotorua have year-round business, conference, and short-break travel. Where the domestic market is itself a summer holiday market, no buffer appears, and the two troughs reinforce each other.

These are associations rather than causes. The warehouse holds no trip-purpose data, so the ski-season explanation is inferred from the seasonal shape rather than confirmed. The divergence between visitor numbers and spending is, however, directly observed.

**Business relevance:** Visitor counts alone are a poor measure of domestic tourism's economic contribution. Fewer but higher-value domestic visitors can sustain revenue when arrivals fall, so an RTO monitoring only arrival numbers may misread its own resilience.

### 5.3 Query 3: RTO dependence on peak-season and international demand

Because domestic demand does not consistently offset periods of weaker international demand, the next question is which RTOs are most exposed to tourism shocks. Query 3 assesses this exposure by measuring each RTO's dependence on international visitor spending and on a small number of peak spending months.

#### *Logic*

Monthly domestic and international spending was first aggregated for each RTO across the complete 2024–2025 period, then averaged by calendar month across the two years. International dependence was measured as the share of total tourism spending generated by international visitors:

```
100.0
* SUM(COALESCE(AvgInternationalSpend, 0))
/ NULLIF(SUM(AvgTotalSpend), 0)
AS InternationalSpendSharePct
```

For example, an international spending share of 60% means that international visitors account for approximately 60% of the RTO's total tourism spending.

Peak-season dependence was measured by ranking each RTO's 12 calendar months from highest to lowest average spending:

```
ROW_NUMBER()
OVER (
  PARTITION BY LocationKey
```

```
ORDER BY AvgTotalSpend DESC
) AS PeakRank
```

The share of total spending generated during the three highest-spending months was then calculated:

```
100.0
* SUM(
  CASE
    WHEN PeakRank <= 3
    THEN AvgTotalSpend
    ELSE 0
  END
)
/ NULLIF(SUM(AvgTotalSpend), 0)
AS Peak3MonthSpendSharePct
```

A higher value indicates that tourism spending is more concentrated in a small number of peak months.

Finally, the two indicators were given equal weight to create a composite vulnerability score:

```
ROUND(
  (
    InternationalSpendSharePct
    + Peak3MonthSpendSharePct
  ) / 2.0,
  1
) AS VulnerabilityScore
```

A higher score therefore indicates greater combined dependence on international demand and peak-season spending. The score is a project-defined comparative indicator rather than an official measure of economic vulnerability.

### ***Result and interpretation***

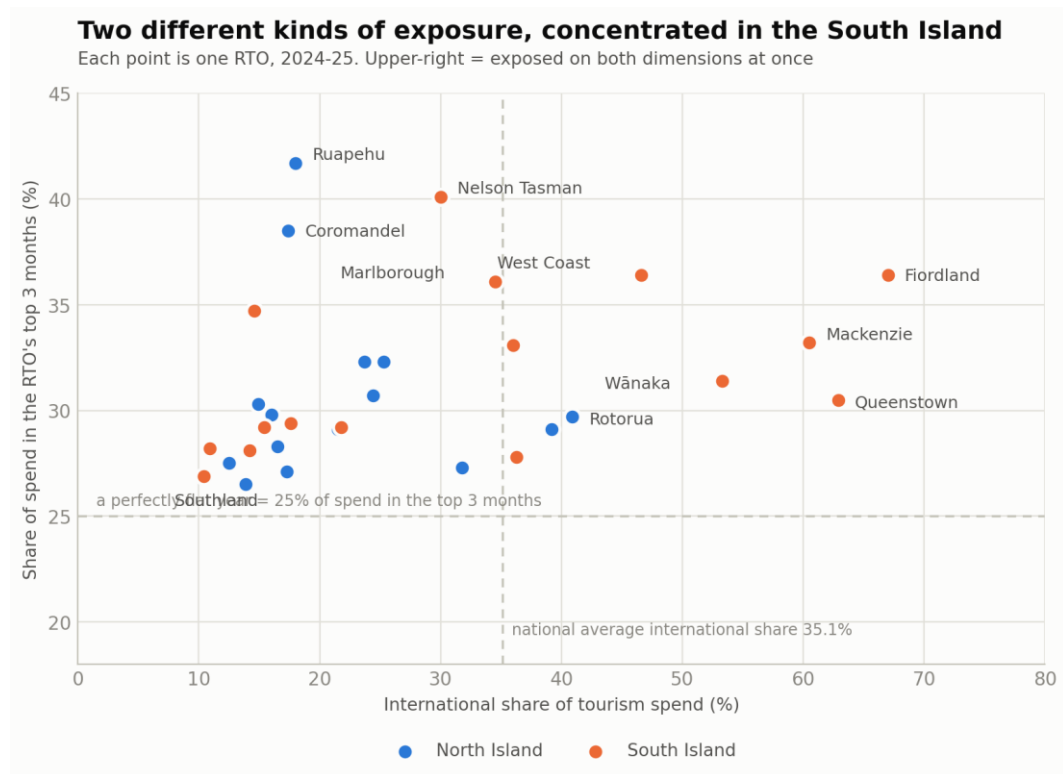


Figure 6. International dependence against peak concentration, one point per RTO.

Points to the **upper right** carry both risks.

Figure 6 also shows that the two risks are genuinely distinct. Fiordland, Mackenzie, and Queenstown sit well to the right, being heavily international but not unusually concentrated. Nelson Tasman, and Coromandel sit high to the left, being only moderately international yet highly seasonal. Ruapehu is the most peak-concentrated RTO in the country at 41.7%, while only 18% international. A ranking based on a single indicator would misclassify all three.

| RTO           | Island | International share | Top 3 months | Score |
|---------------|--------|---------------------|--------------|-------|
| Fiordland     | South  | 67.0%               | 36.4%        | 51.7  |
| Mackenzie     | South  | 60.5%               | 33.2%        | 46.8  |
| Queenstown    | South  | 62.9%               | 30.5%        | 46.7  |
| Wānaka        | South  | 53.3%               | 31.4%        | 42.4  |
| West Coast    | South  | 46.6%               | 36.4%        | 41.5  |
| Rotorua       | North  | 40.9%               | 29.7%        | 35.3  |
| Marlborough   | South  | 34.5%               | 36.1%        | 35.3  |
| Nelson Tasman | South  | 30.0%               | 40.1%        | 35.1  |
| Kaikōura      | South  | 36.0%               | 33.1%        | 34.6  |
| Auckland      | North  | 39.2%               | 29.1%        | 34.1  |

Table 5. Most exposed RTOs by composite vulnerability score.

Figure 6 and Table 5 both show exposure is concentrated in the South Island, which holds 8 of the 10 highest-scoring RTOs. Its average international share is 33.3%, against 22.2% in the North Island. Fiordland is the

extreme case, earning two-thirds of its tourism income from overseas visitors and more than a third of it within three months.

One caveat applies. The vulnerability score is a project-defined exploratory measure, not an official MBIE indicator. The two components are weighted equally because there is no empirical basis for preferring one over the other, so the ranking should be read as a screening tool rather than a definitive risk assessment.

**Business relevance:** The two types of exposure call for different responses. RTOs high on international dependence benefit from domestic market development, while those high on peak concentration benefit from off-season products, events, and campaigns. Reading the two axes separately lets an agency match the response to the actual risk.

## 6. Insights and Future Work

### 6.1 Key insights

#### *Seasonality is a geography, not just a calendar*

Both islands peak in summer, but the South Island swings 53 index points between its best and worst month, against 34 in the North. Seven of the ten most seasonal RTOs are in the South Island. What separates the steady regions from the volatile ones is economic diversity rather than tourism scale. Wellington and Waikato hold up because part of their tourism spending is business and events demand that ignores the holiday calendar. Fiordland and Ruapehu rise and fall with a single attraction.

#### *Domestic demand stabilizes through value, not volume*

Domestic tourism does not offset weaker international demand through higher visitor volumes. Domestic visitor numbers fall in every RTO during low international-demand months, with a median decline of 19.9%. Spending is more resilient: domestic spending falls by a median of only 5.6% and rises in 11 of 31 RTOs, while international spending falls in all 31 by a median of 47.4%.

This suggests that domestic tourism supports regional income mainly by providing a relatively stable spending base rather than by filling the off-season gap. Ruapehu is the clearest exception, with domestic spending rising by about 111%, reflecting its winter snow market. A strong domestic buffer therefore appears to depend on having a distinct off-season demand driver.

#### *The two risks compound rather than offset*

This finding ties the project together. The RTOs identified in Query 3 as most exposed, including Fiordland, Nelson Tasman, Marlborough, and West Coast, are largely the same ones Query 2 shows have no domestic offset. International dependence and the absence of a domestic buffer are not independent risks. The regions that would most need domestic demand during a shock are precisely those least able to draw on it.

#### *Per-visitor value explains why volume shocks hurt*

International visitors spend roughly \$269 each in the North Island and \$192 in the South Island, compared with \$93 and \$96 for domestic visitors. A modest fall in international arrivals, therefore, removes far more income than an equivalent fall in domestic arrivals. Regions cannot substitute headcount for headcount.

**Equity implication.** A downturn in international arrivals would not be felt evenly. It would fall hardest on small South Island communities where tourism accounts for a large share of local employment, and there is no alternative demand to cushion the blow. These are also the regions with the least economic diversification to absorb the loss, so the burden falls on the places least able to carry it.

## 6.2 Future work

The most valuable extension is the origin of tourists. MRTE records spending across 29 origin markets and 7 product categories, but the current fact table sums them both away because ADP and TV&F have no equivalent breakdown. Rather than widening the existing fact table, a second fact table - FactSpendDetail at Month, RTO, Product, and Origin would share **DimDate** and **DimLocation** as conformed dimensions.

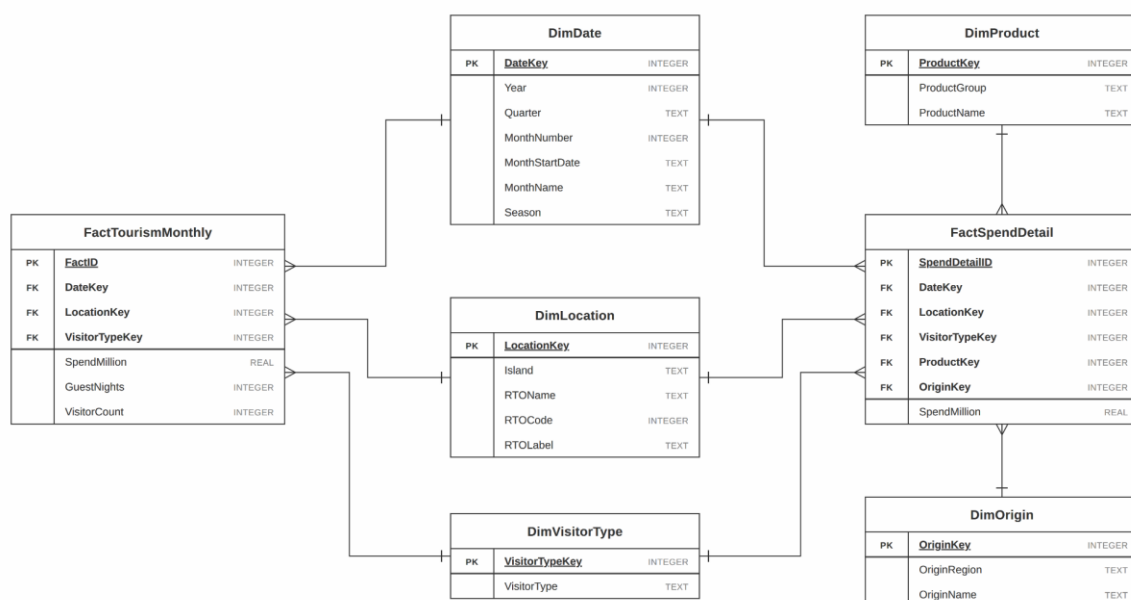


Figure 7. Proposed extension: a second fact table - FactSpendDetail - sharing the existing conformed dimensions.

A prototype star schema shown in Figure 7 confirms that the design works, since the detailed fact reconciles exactly with the current fact for total spending. Early results also show its value. The United States is New Zealand's largest international tourism earner over the two years, ahead of Australia, and the top three markets together account for 64% of international spending. That concentration considerably sharpens the exposure finding.

Three smaller improvements would also strengthen the analysis. Extending the series back through MBIE's Tourism Electronic Card Transactions would separate structural patterns from post-pandemic recovery. Retaining the 'Local resident' counts already present in TV&F would allow a visitors-per-resident ratio, the standard measure of tourism pressure on communities. Finally, a sensitivity analysis of the vulnerability weighting would show whether the top-ranked RTOs are robust or artifacts of the equal-weight assumption.

## References

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**Note on terminology.** MRTE measures visitor expenditure only, with no costs, wages or margins, so this report refers to the highest-earning categories and regions rather than the most profitable.

**Note on GDP figures.** Tourism's 7.7% share of GDP combines the direct and indirect contribution. The direct contribution alone was \$18.0 billion, or 4.6% (Tourism New Zealand, 2026).

## Appendix A: Submitted files

### 1. Source datasets

- a. [MRTE — Monthly Regional Tourism Estimates](#)
- b. [ADP — Accommodation Data Programme](#)
- c. [TV&F — Tourism Volumes and Flows](#)
- d. [Stats NZ RTO Areas 2025](#)

### 2. [Project\\_702\\_Data\\_Cleaning.sql](#) – Cleans and transforms the four raw source tables into structured, typed, and constrained tables ready for warehouse development.

### 3. Cleaned database

- a. [Project\\_702\\_Clean.db.sql](#) – Contains the cleaned source tables produced after the data-cleaning stage.

### 4. [Project\\_702\\_DataWarehouse\\_and\\_Analysis.sql](#) – Builds and populates the star schema, creates supporting indexes, performs data validation, and includes the SQL queries used to answer the three research questions.

### 5. Final database

- a. [Project\\_702\\_Final.db.sql](#) – Contains the complete cleaned, transformed, and dimensional data warehouse, ready for analytical querying and reporting.

## Appendix B: Visualization

### Dashboard: [Streamlit Dashboard](#)

The interactive visualisation is provided through a Streamlit web application. It presents key analytical results from the final database, allowing users to explore the data, compare regional patterns, and interact with the visualisations.